

Nicholas Eisenberg, Ph.D.

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Experience

Senior Data Scientist

July 2022 - Present

Nevada National Security Site

- Built and managed a MySQL database of seismic sensor data using SQLAlchemy, optimizing data retrieval for analysis purposes
- Acted as the lead investigator for project involving the organization and feature engineering of a dataset consisting of 10,000+ shots from the Los Alamos National Lab linear accelerator
- Constructed a predictive model using Keras to forecast future performance of an X-ray machine with accuracy of %90, resulting in improved maintenance planning and operational efficiency
- Developed interactive dashboards using Streamlit, Plotly, and Bokeh with Holoviews to visually present complex data and provide actionable insights to fellow scientists and stakeholders
- Applied Scikit-learn and Scikit-image to analyze and characterize the spot of an X-ray machine, thus estimating the blur kernel for deblurring X-ray images, enhancing image quality, and aiding in accurate analysis
- Acted as lead maintainer of a company-wide internal Python library hosted on our internal GitLab, ensuring code quality, version control, and collaboration among team members

Graduate Teaching Assistant

January 2016 - June 2022

Auburn University and University of Nevada Las Vegas

- Instructed courses in College Algebra and Pre-Calculus
- Instructed weekly recitation sections for Mathematical Statistics, Real Analysis and Calculus I, II and III
- Managed Auburn University's Math tutoring center of 14 tutors.

Technical Skills

Programming Languages: Python, SQL, R, Lua, Bash, $\text{\LaTeX}$

Libraries: scikit-learn, tensorflow, keras, pytorch, scipy, plotly, streamlit, pymysql, sqlalchemy

Developer Tools: Git, Unix/Linux, Neovim, AWS

Publications

- 1: Chen, L., Eisenberg, N. Interpolating the stochastic heat and wave equations with time-independent noise: solvability and exact asymptotics. Stoch PDE: Anal Comp (2022). <https://doi.org/10.1007/s40072-022-00258-6>
- 2: Chen, L., Eisenberg, N. Invariant measures for the stochastic heat equation on $\mathbb{R}^d$ without drift term. Submitted January 2023 to Journal of Theoretical Probability. Preprint available at <https://arxiv.org/pdf/2209.04771v1.pdf>

Education

Auburn University, Ph.D.

June 2022

Research Area: Probability (Stochastic Processes and Partial Differential Equations)

Advisor: Le Chen

University of Nevada Las Vegas, B.S./M.S.

December 2015 / January 2018

Major: Applied Mathematics